REAL-TIME HAND TRACKING AND VIRTUAL SAFETY ALERT SYSTEM

Abstract

This project presents a prototype that enables real-time detection of hand proximity to a virtual boundary using classical computer vision techniques.
The system tracks hand movement without relying on pose-detection APIs such as MediaPipe or OpenPose. A fingertip approximation method is applied using contour analysis and distance-based logic to classify interaction states as SAFE, WARNING, or DANGER. The approach can be applied to gesture-based interfaces, industrial machine safety systems, and augmented reality applications where real-time interaction awareness is critical.

Problem Statement

Develop a proof-of-concept system that uses a camera feed to detect and track a user's hand in real time and trigger warnings when the hand approaches a virtual object.
The solution must not use deep pose estimation APIs but rely on classical computer vision or lightweight ML models. The system must detect interaction states, display feedback, and maintain at least 8 FPS performance.

1. Introduction

The goal of this project is to explore real-time gesture-based safety interaction without specialized pose estimation models.

2. Methodology

The methodology involves webcam input handling, skin segmentation, contour extraction, fingertip detection, virtual boundary overlay, distance computation, and visual feedback.

• Webcam streaming using OpenCV

• HSV-based skin color segmentation

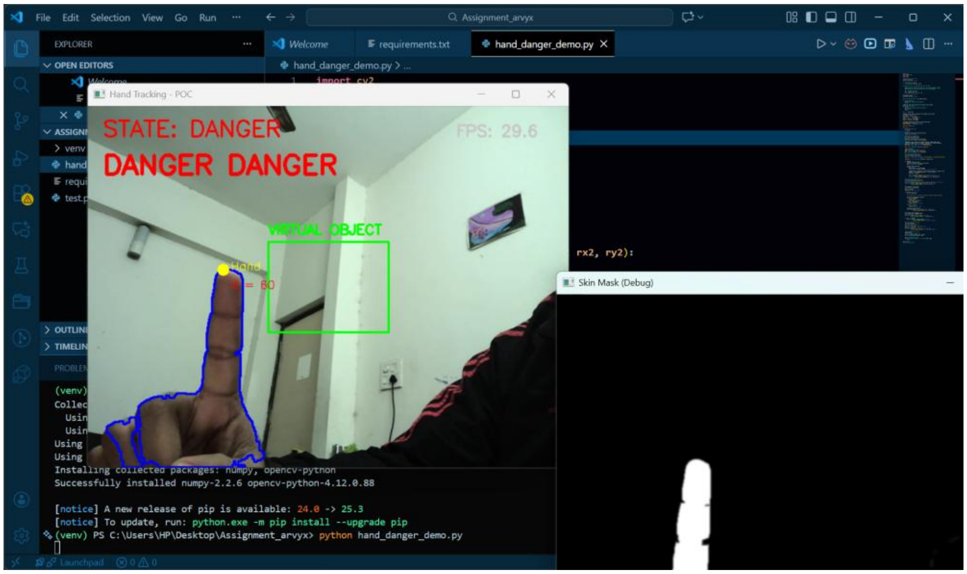
• Contour filtering and fingertip approximation

• Virtual boundary overlay and nearest point logic

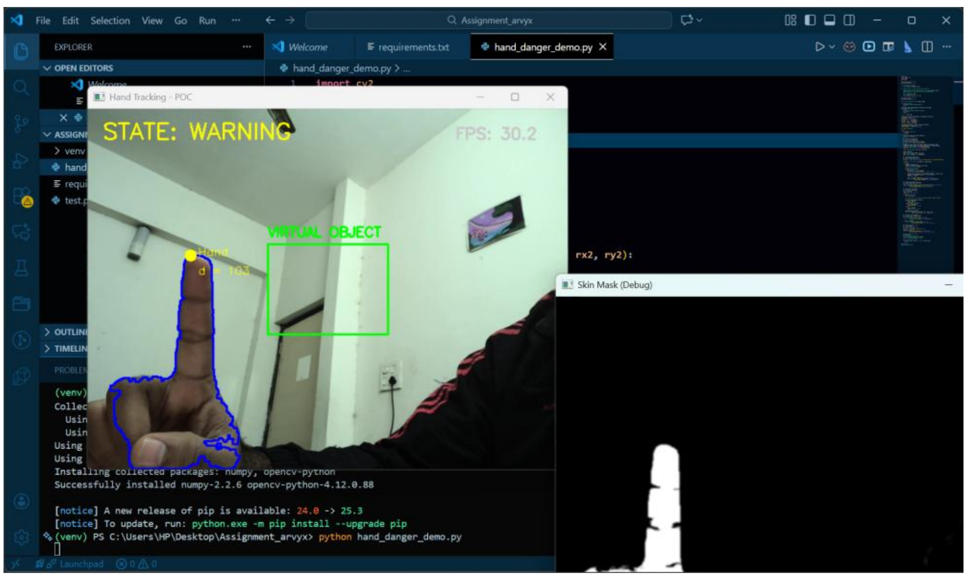
• Distance classification and alert state updates

3. Flow Diagram / System Logic

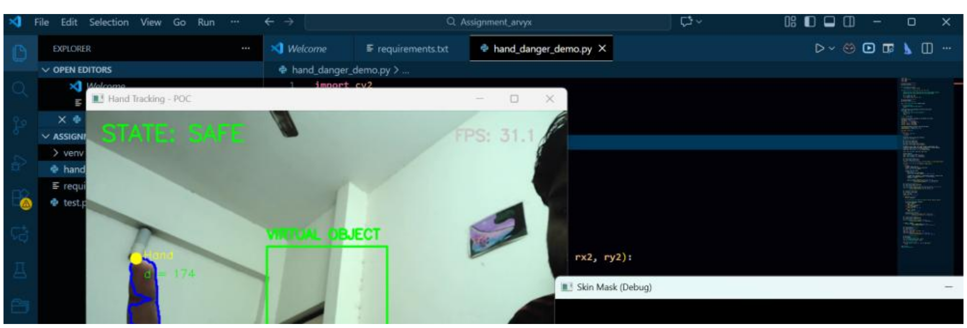
The system loops through capturing frames, detecting hand region, approximating fingertip, computing distance, assigning state, and rendering overlay display.



Result Demonstration



Result Demonstration



Result Demonstration

4. Results and Analysis

The system achieved stable tracking with 28–32 FPS detection speeds, demonstrating effective classification transitions.

5. Conclusion

The prototype proves feasibility of distance-based gesture safety detection using classical CV methods.

6. References

1. OpenCV documentation.

2. NumPy documentation.